

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2022

Course Code: CIV102

Course: ENGINEERING MECHANICS

Question Paper No. :UGF045

Max. Marks:100

PART – A

Answer all the questions

10 x 2 = 20 Marks

1. For the given system, evaluate the moment produced by T about the origin O . $OA = 15$. All dimensions are in m. Figure 1.

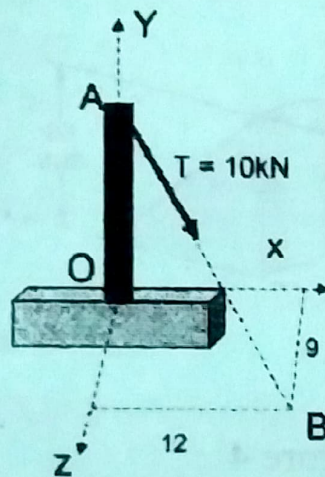


Figure 1

2. The top view of a revolving entrance door is shown. Two persons simultaneously approach the door and exert force of equal magnitudes. If the resulting moment about the door pivot axis at O is 25 Nm, determine the force magnitude F . Figure 2.

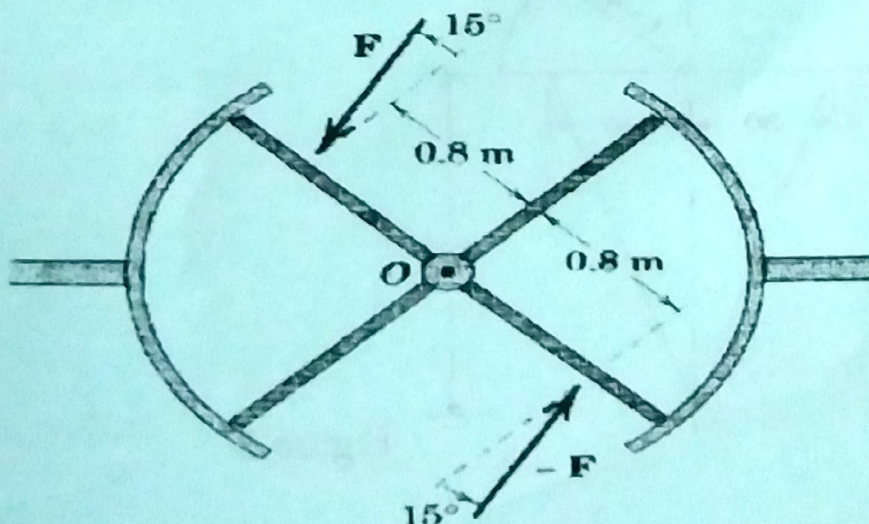


Figure 2

3. For the given wire, locate the coordinates for the centroid. Figure 3.

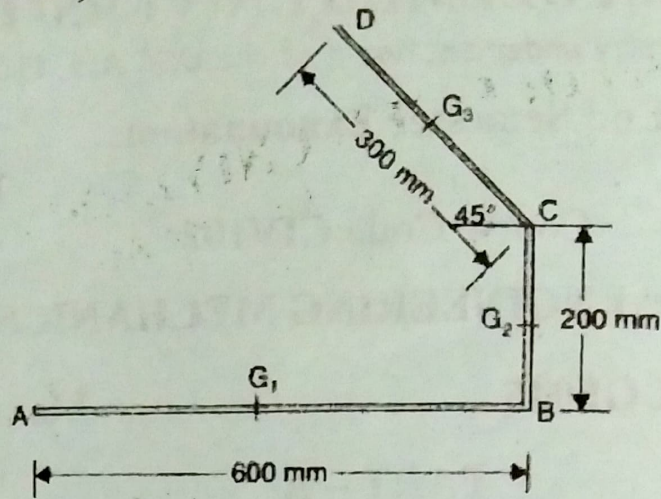


Figure 3

4. If the right triangular area is revolved through 180° about the z-axis, calculate the volume of solid generated. Figure 4.

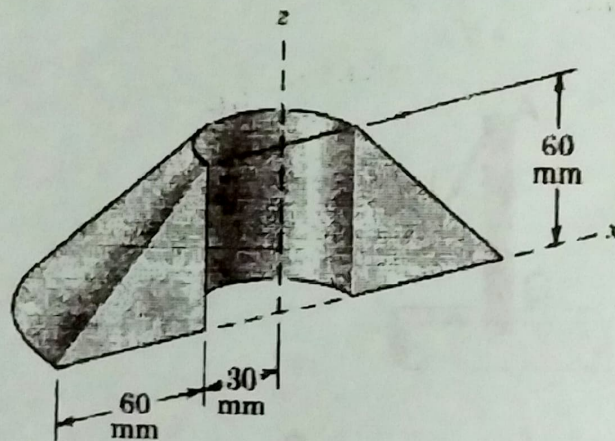


Figure 4

5. Evaluate the moment of inertia of the section about the symmetrical axis. Figure 5.

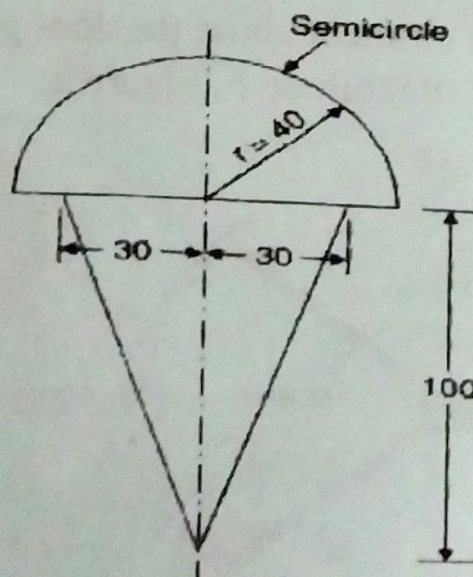


Figure 5

6. The 12 N wooden block is used for level support of the 90 N can of paint. Evaluate the magnitude and direction of the friction force exerted by the roof surface on the wooden block. Figure 6.

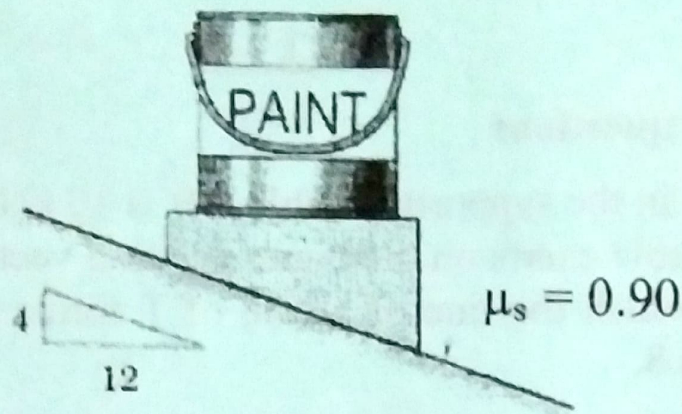


Figure 6

7. The 40 mm diameter screw has a double square thread with a pitch of 12 mm and a lead of 24 mm. The coefficient of friction between the contact surfaces is 0.15. If a torque $M = 60$ Nm is applied as given in the figure, determine the force P which would allow the shaft to move to left at a constant speed. Figure 7.

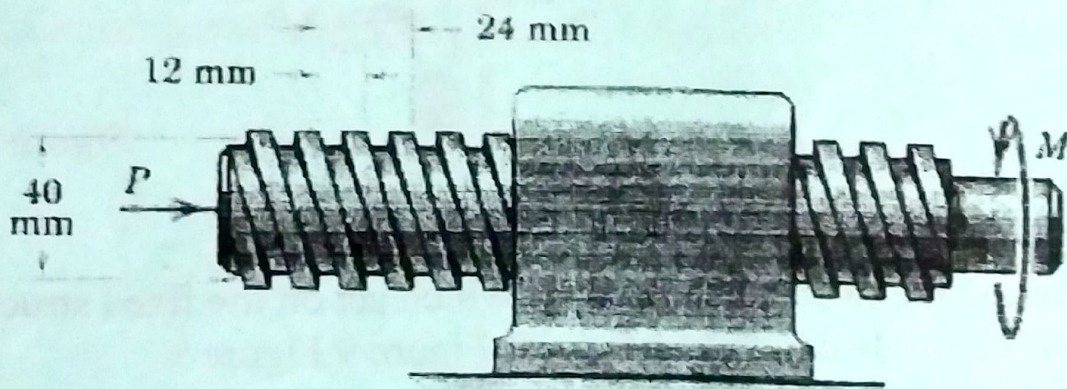


Figure 7

8. On turning a corner, a motorist rushing at 15 m/s, finds a child on the road 40 m ahead. He instantly applies the brake to stop the vehicle within 5 m from the child. Calculate the retardation and time required to stop the vehicle.
9. An elevator cage of mass 900 kg is lifted or lowered vertically by means of a rope. A person of mass 72.5 kg entered the cage and the cage started to move up with an acceleration of 3 m/s^2 . Arrive at the force exerted by the man on the cage.

10. Two railway trucks of masses 10000 kg and 8000 kg moving in the same direction of speeds 10 m/s and 15 m/s respectively collide and subsequently move together. Evaluate the common speed due to impact.

PART – B

Answer all the questions

4 x 15 = 60 Marks

11. The tension in the supporting cable AB is 10 kN. Arrive at the force which the cable exerts on the boom BC as a vector T . Also calculate the angles which the line of action of T forms with the coordinate axes. Figure 8.

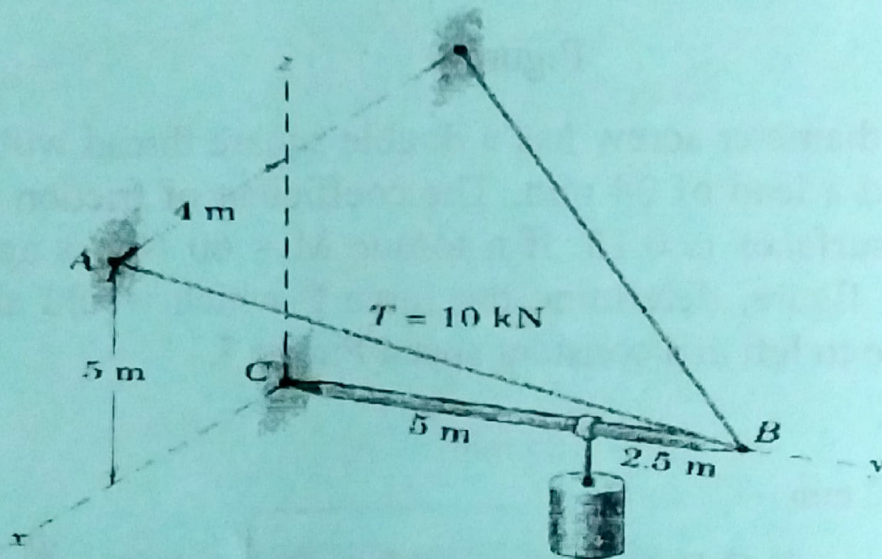


Figure 8

(OR)

12. Combine the two forces P and T , which act on the fixed structure at B , into a single equivalent force R Figure 9. Figure 9

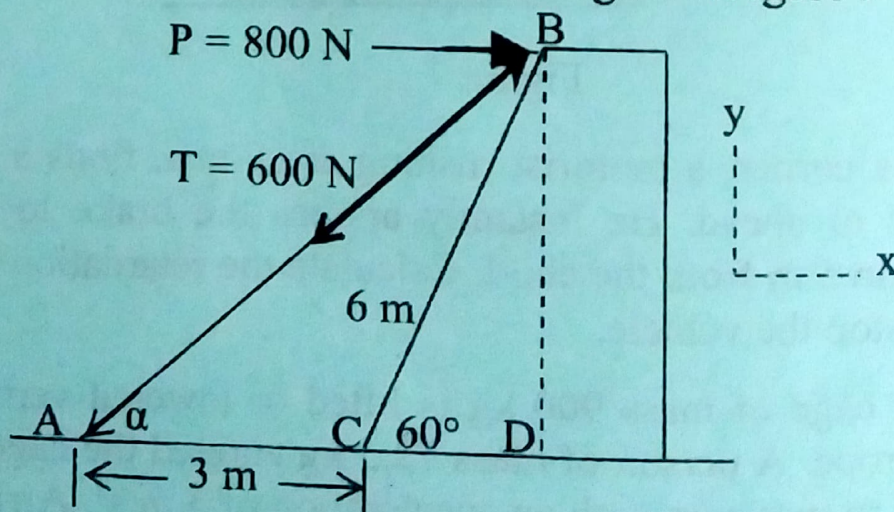


Figure 9

13. Locate the x and y coordinates of the centroid of the shaded area. Figure 10.

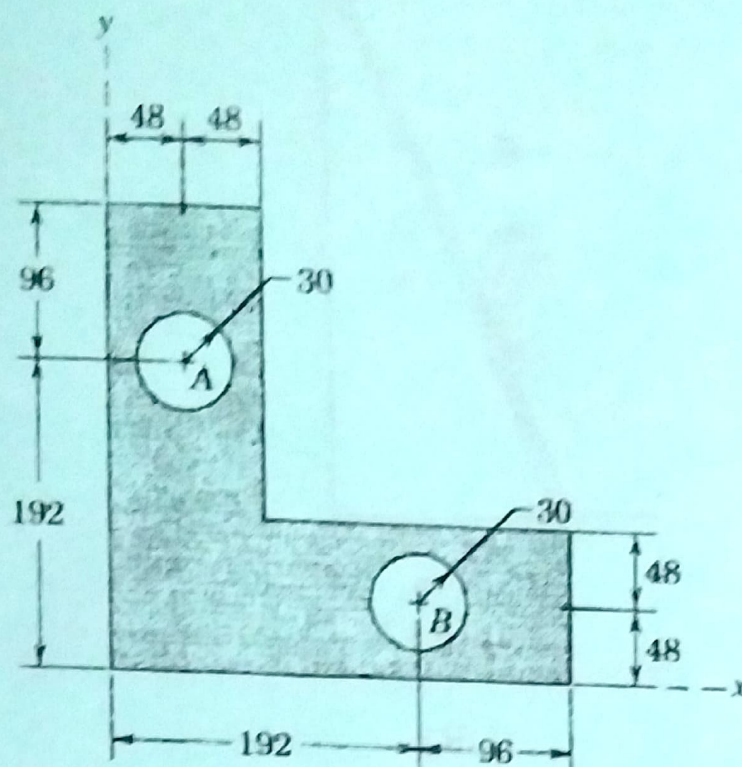


Figure 10

(OR)

14. The shaded area is the cross section of bearing block. Evaluate the moment of inertia about the base a-a. Figure 11.

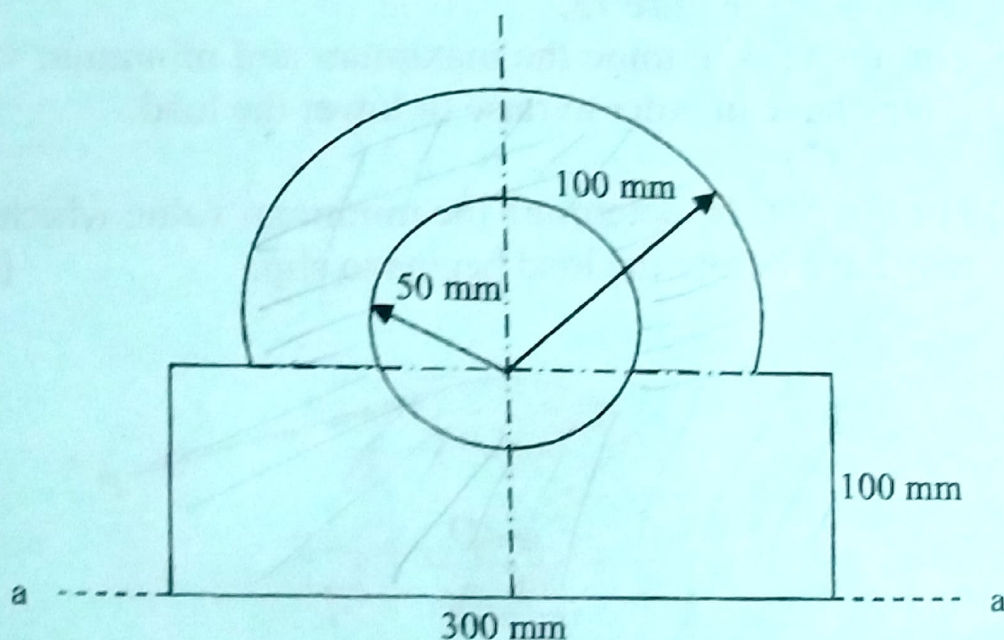


Figure 11

15. Evaluate the distance ' s ' to which the 90 kg painter can climb without causing the ladder to slip. The top of the 15 kg ladder has a small roller, and at the ground the coefficient of static friction is 0.25. Figure 12.

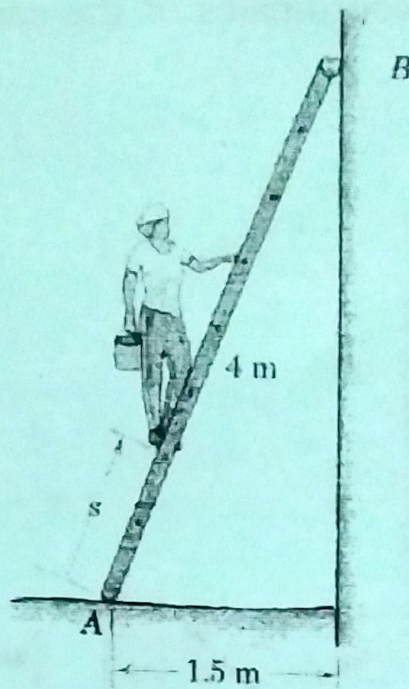


Figure 12

(OR)

16. A flexible cable which supports the 100 kg load is passed over a fixed circular drum and subjected to a force P to maintain equilibrium. The coefficient of static friction between the cable and the drum is 0.3. Figure 13.

(i) For $\alpha = 0$, determine the maximum and minimum values which P may have in order to raise or lower the load.

(7)

(ii) For $P = 500$ N, determine the minimum value which the angle α may have before the load begins to slip.

(8)

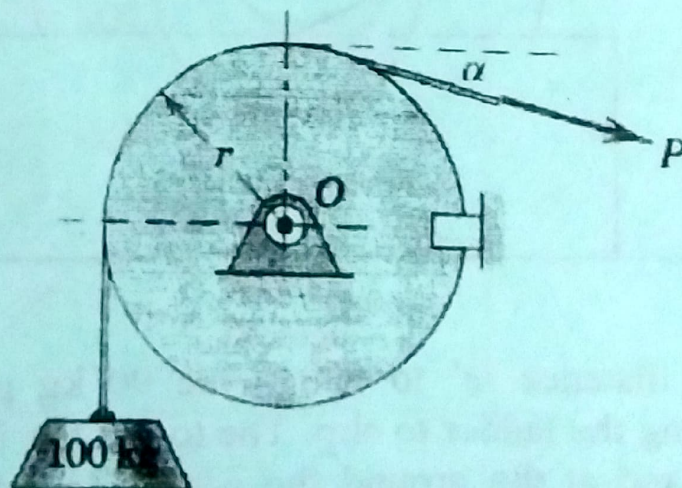


Figure 13

17. In a racing event the rider jumps off the slope at 30° , from a height of 1 m and remained in midair for 1.5 s. Determine the speed at which he was traveling off the slope, the horizontal distance he travels before striking the ground, and the maximum height he attains. Neglect the size of the bike and rider. Figure 14.

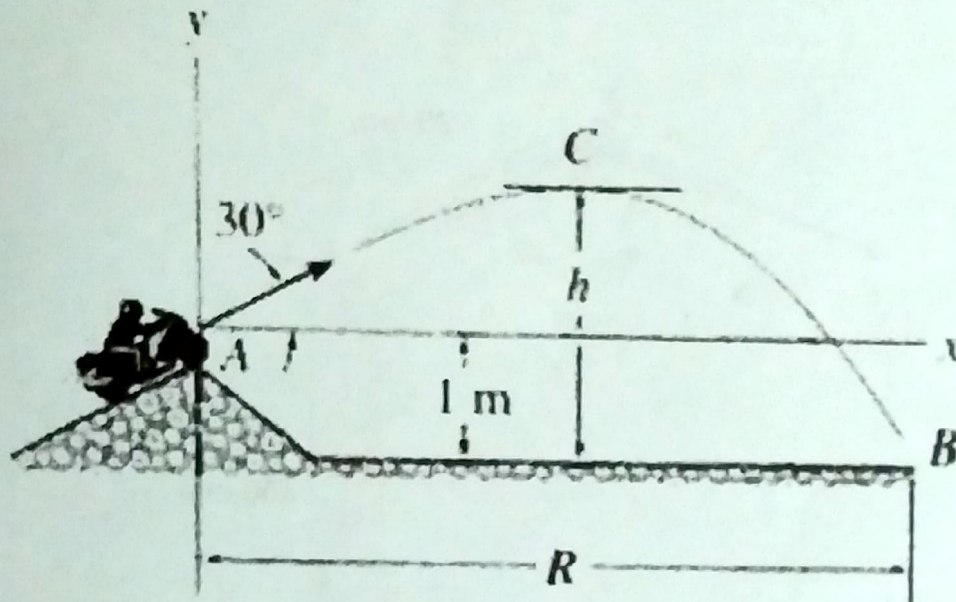


Figure 14

(OR)

18. A pile of mass 1000 kg is driven vertically into the ground by the blows of the pile hammer of mass 2000 kg and falling on the head of the pile from a height of 1 m, in each blow. If the average resistance of the ground is 200 kN evaluate (i) pile penetration per blow (ii) duration for which the pile is in motion. Figure 15.

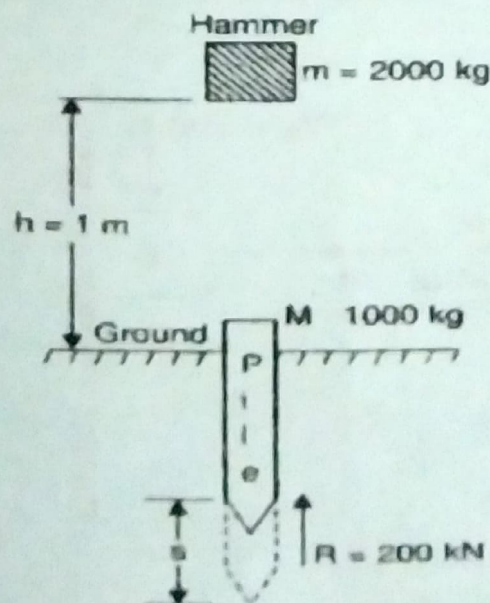


Figure 15

PART - C

Answer the following

1 x 20 = 20 Marks

19. (i) The 180 N force is applied to the end of body OAB. If $\theta = 50^\circ$, determine the equivalent force-couple system at the shaft axis O. Figure 16. (5)

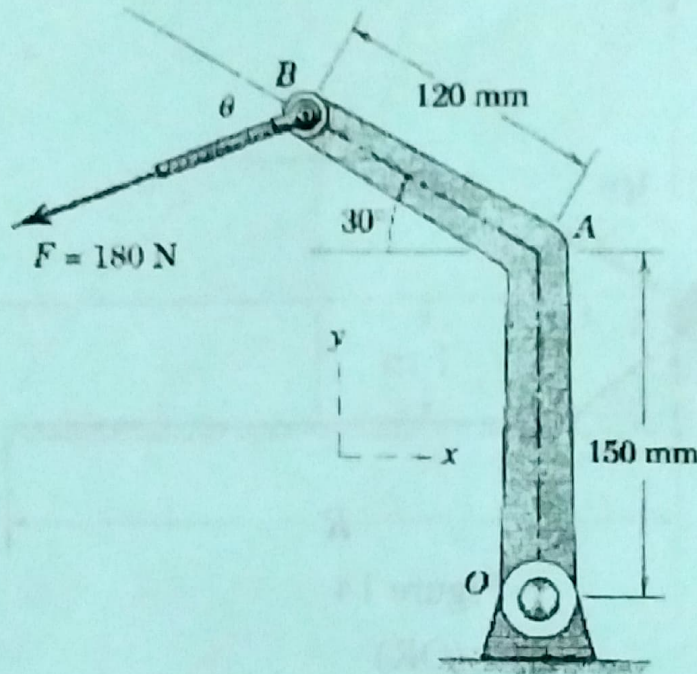


Figure 16

- (ii) A block A is placed on an inclined plane and held in position by means of a horizontal rod hinged to block B. If $W_B = 500\text{ N}$, find the minimum value of W consistent with equilibrium. The coefficient of friction $\mu_A = 0.2$ and $\mu_B = 0.4$. Figure 17. (15)

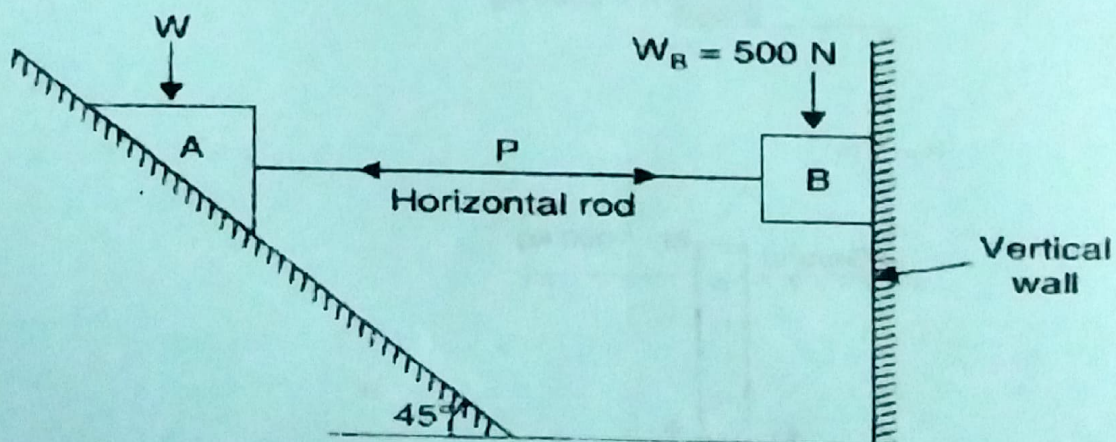


Figure 17

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